

Multi-modal Retrieval-Augmented Generation (RAG): Tackling Hallucination in Multimodal Question Answering

LAM, Ching Yin

DAN1

Advised by Prof. Dan XU
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Department of Computer Science and Engineering
The Hong Kong University of Science and Technology

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Outline

- 1 Introduction
- 2 Methodology
- 3 Testing and Evaluation
- 4 Discussion
- 5 Conclusion

Overview of RAG

- Retrieval-Augmented Generation (RAG) has become the dominant method in grounding LLM generation with evidence
- Successful in text-only settings

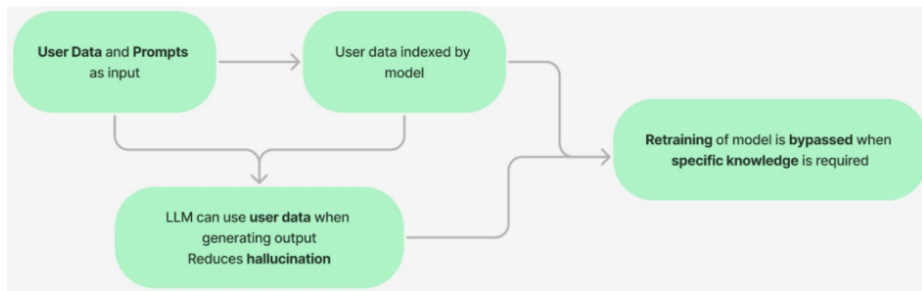


Figure: How RAG benefits LLM generation

Why Multimodal RAG?

Multimodal RAG can be useful in high-stakes domains and situations, such as

- a domain expert querying a MLLM on fine differences between rare bird species
- an engineer asking a MLLM about specific design elements in blueprints
- a doctor inspecting X-ray scans for certain diseases with MLLM assistance

Problem and Objectives

The problems with current RAG pipelines are:

- Current text-only RAG pipelines do not apply to multimodal LLMs (MLLMs)
- Limitations in emerging multimodal RAG pipelines:
visual cues or fine details overlooked by visual encoders (such as CLIP) used
- Grounding with retrieved visual evidence remains challenging

This thesis is built on the following goals and hypothesis:

- **Hypothesis:** separate feature spaces for visual and text outperforms unified embeddings
- Preserve fine-grained visual features in visual evidence retrieval
- Use state-of-the-art specialized models to enable cross-modal retrieval without fine-tuning

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High-Level Architecture

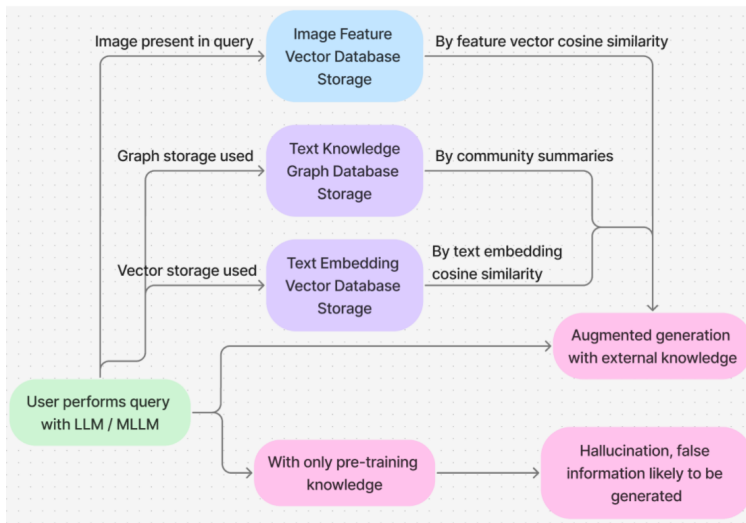


Figure: A high level overview of the entire project and implementation

Document Preprocessing

- Split document text into fixed-sized overlapping text chunks
- Extract inline images and surrounding text context for captioning
- Replace images with their captions

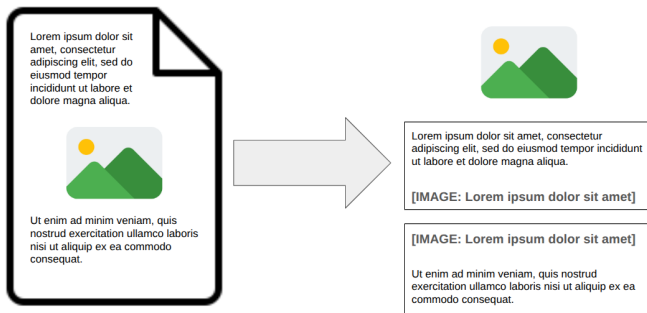


Figure: Our document parsing method

Embedding

We embed text and extracted images separately.

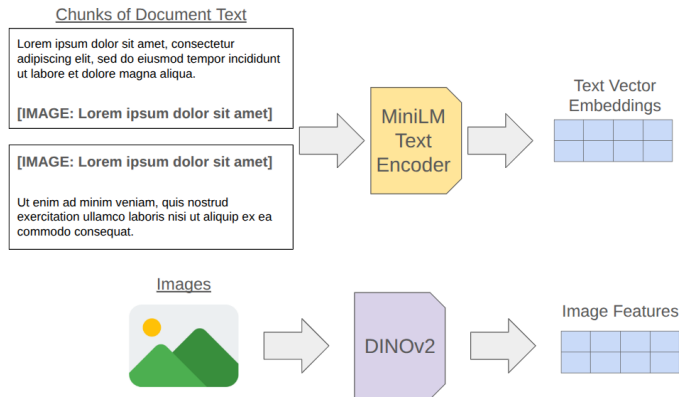


Figure: Embedding text and images into separate representations

Embedding

For **text**,

- Map sentences or paragraphs into high-dimensional dense vector space
- Use MiniLM model (sentence-transformers/all-MiniLM-L6-v2) to embed text chunks

For **images**,

- Use DINOv2, a vision-specific model, to extract feature vectors
- Embed captions in the same way as text chunks

Storage

We store text embedding vectors and image feature vectors separately.

- **Text embeddings** stored in a vector database
- **Image features** and **caption text embeddings** stored in a *multi-vector* database

Alternately, we also experiment with knowledge graph storage.

- Extract **entities** and **relationships** with an LLM
- Entities form nodes, relationships form edges
- Detect communities with Louvain's Algorithm
- Generate community summary for each community

Processing Flow

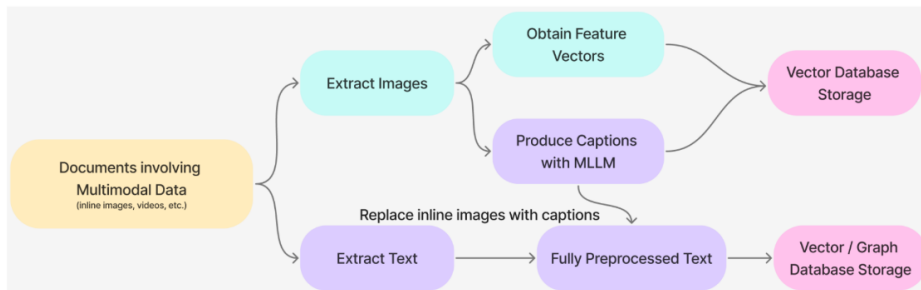


Figure: Flow of embedding and storing documents

Retrieval and Generation

Query enhancement using the **Hypothetical Document Embedding (HyDE)** method:

- Generate a **zero-shot answer** to the raw user query
- Embed and use the zero-shot output for retrieval
- Zero-shot answer contains semantic nuances and text patterns of expected document chunks

However, ablation test results show no significant advantage. Left as a configurable option.

Retrieval and Generation

Queries to a MLLM may be text-only, or involve text and images.

- **Text-to-text search:** retrieve similar text chunks by text embedding vector cosine similarity
- **Image-to-image search:** obtain DINOv2 features for query images if present, retrieve similar images by feature vector cosine similarity
- **Text-to-image search:** embed query text into DINOv2 feature space using Talk2DINO, retrieve images by cosine similarity

Talk2DINO uses the vanilla CLIP text encoder with a limited context size of 77 tokens. We use Long-CLIP before the Talk2DINO projection layer for longer context of up to 248 tokens.

Retrieval and Generation

Top k retrieved document chunks and images are used as context in generation.

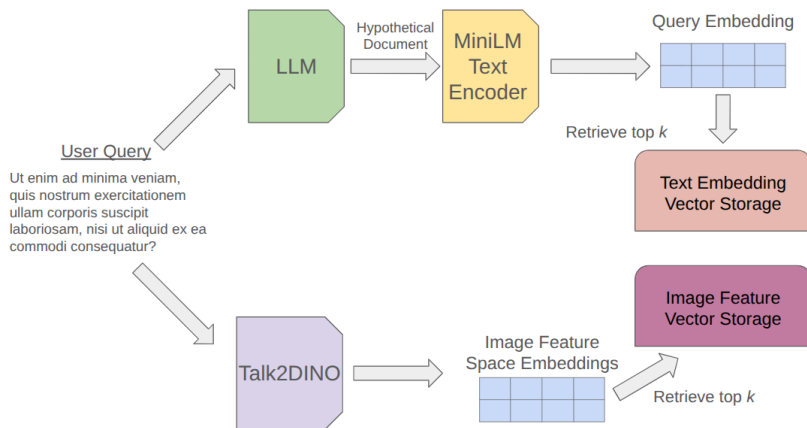


Figure: Retrieval by text-to-text and text-to-image

Retrieval and Generation

When graph storage option is used, text-to-text retrieval goes through the following steps:

- Embed query with MiniLM text embedding model
- Embed community summaries with MiniLM text embedding model
- Compare relevance of community using embedding vector cosine similarity

The community summaries of top k most relevant communities are used as retrieved context in generation.

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Testing

We perform a small-scale test on a corpus of 4 documents retrieved from Wikipedia. We obtain zero-shot, text-only retrieval, and multimodal retrieval accuracy scores using GPT-4o-mini.

- The level of accuracy increases from zero-shot to text-only retrieval, then multimodal retrieval.
- The highest accuracy score is at $k=7$ and 10 under multimodal retrieval.

This verifies that retrieval benefits answer generation.

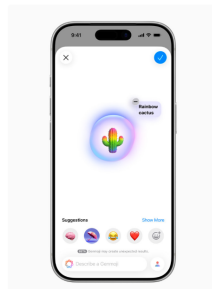
Test Example

The following is a screenshot taken from an example document in our test corpus.

Create Genmoji to Fit Any Moment

Users can create their own unique emoji with the power of Apple Intelligence, making conversations with family and friends more fun and playful. Users can type a description to create an emoji. They can also mix together multiple emoji, or combine emoji and text to create something entirely new. With images from their photo library, users can take Genmoji even further by creating one that is inspired by a friend or family member. And similar to Image Playground, users can also change expressions or modify personal attributes to match their contact's

latest look.



Users can create their own unique emoji with Genmoji, making conversations with family and friends more fun and playful.

Figure: Part of a document from the test corpus

Test Example

For the question:

What is Genmoji like and how can I use it? Can you give me an example?

Multimodal

...

For example, you can create a "rainbow cactus" emoji by describing it, and then you can further customize it by ...

Text-only

...

For example, if you wanted to create an emoji inspired by your friend who loves cats, you might start by ...

Zero-shot

1. Access the Platform: Visit the Genmoji website or app.
2. Select Options: Choose facial expressions, accessories, hairstyles...

Multimodal retrieval mode uses the “rainbow cactus” visual context, while text-only retrieval cannot. We also see hallucinations in the zero-shot answer, such as a non-existent “website”.

Benchmarking on MMDocRAG

MMDocRAG is a document retrieval and generation benchmark with a dataset of 222 long documents and 4,055 expert-verified QA pairs.

Our results for **text-only retrieval** are as follows.

Model	GPT-4o-mini	Qwen2.5VL
Pure-text input sequence to MLLM		
Zero-shot (no retrieval)	49.9 (–)	69.2 (–)
5 quotes (Top- k , $k=5$)	50.4 (–)	59.6 (–)
7 quotes (Top- k , $k=7$)	51.1 (–)	59.8 (–)
10 quotes (Top- k , $k=10$)	54.0 (–)	61.0 (–)
15 quotes (Top- k , $k=15$)	56.9 (70.0)	61.8 (61.2)
20 quotes (Top- k , $k=20$)	57.4 (69.8)	62.0 (61.4)

Benchmarking on MMDocRAG

MMDocRAG is a document retrieval and generation benchmark with a dataset of 222 long documents and 4,055 expert-verified QA pairs.

Our results for **multimodal retrieval** are as follows.

Model	GPT-4o-mini	Qwen2.5VL
Pure-text input sequence to MLLM		
Zero-shot (no retrieval)	49.9 (–)	69.2 (–)
Multimodal input sequence to MLLM		
5 quotes (Top- k , $k=5$)	52.7 (–)	65.8 (–)
7 quotes (Top- k , $k=7$)	55.6 (–)	66.4 (–)
10 quotes (Top- k , $k=10$)	55.6 (–)	67.0 (–)
15 quotes (Top- k , $k=15$)	57.5 (66.7)	66.4 (63.2)
20 quotes (Top- k , $k=20$)	55.5 (64.6)	65.8 (62.6)

Benchmarking on MMDocRAG

- GPT-4o-mini lags behind, while Qwen2.5VL outperforms original paper
- Possibly due to performance differences between Azure-provided and OpenAI models
- Factuality scores show increasing trend with larger k

Benchmarking on Visual-RAG

Visual-RAG is based on the iNaturalist 2021 Competition training dataset of 2.7M images of 10,000 biological species. Due to time and compute limits, we use the test dataset of 100,000 images to simulate this experiment.

Results for end-to-end generation are as follows.

Model	GPT-4o-mini (GPT-4o)	Qwen2.5VL
Zero-shot (no image)	43.05 (49.33)	56.68 (34.84)
Top- k , $k=1$	7.49 (22.65)	37.43 (37.96)
Top- k , $k=3$	11.76 (38.50)	33.42 (39.44)
Top- k , $k=5$	15.24 (43.45)	39.57 (39.84)
Top- k , $k=7$	14.44 (46.52)	40.91 (38.99)
Top- k , $k=10$	17.11 (46.93)	42.25 (42.91)
Top- k , $k=15$	19.25 (47.33)	46.79 (42.78)
Top- k , $k=20$	17.91 (50.00)	43.58 (43.44)

Benchmarking on Visual-RAG

We further analyze retrieval logs from end-to-end generation experiments, and use the following metrics to evaluate retrieval.

- Recall @ k : fraction of ground truth images among the top k retrieved images
- Normalized Discounted Cumulative Gain (NDCG) @ k : ranking quality (rewards correct images with higher ranking, discounts relevance logarithmically with position)
- Hit @ k : probability of at least one clue image among the top k images
- Hit Count @ k : number of clue images found in the top k retrieved images

Benchmarking on Visual-RAG

Our experiment conditions for retrieval analysis are slightly different from the original paper.

- For retrieval evaluation, the original paper authors select 200-300 images per query
- We perform retrieval among all 100,000 images in the test set, making our experiment more difficult for the retriever

Our results still outperform the original paper in most settings despite the increased difficulty.

- Recall @ k , Hit @ k , and Hit Count @ k remain higher than other methods in the paper
- NDCG values decline for $k > 5$ due to limited number of ground truth images

These results suggest that our pipeline allows minor fine-grained details to be considered in retrieval.

Additional Note on Benchmark Results

Zero-shot generation with Qwen outperforms generation with retrieval, but GPT-4o-mini follows the expected trend.

- Post-training with Supervised Fine-Tuning (SFT) on Qwen2.5 gives long-context generation and logical reasoning
- Longer zero-shot answers have higher chance of hitting factual points
- LLM judge may award more points as a result
- Generation with retrieval does not benefit from model characteristics, model prompted to strictly generate based on retrieved context

Ablation Test: CLIP vs DINOv2

We perform an ablation test on using CLIP for embedding images and multimodal retrieval on GPT-4o-mini.

- Our DINOv2-Talk2DINO pipeline achieves higher accuracy for $k=1, 3, 7, 10$ compared to CLIP
- The highest accuracy is from using the DINOv2-Talk2DINO pipeline
- CLIP underperforms DINOv2 in most settings
- Unified representation space for text and vision in CLIP loses fine-grained details
- Results validate the dual representation design for multimodal processing

Ablation Test: HyDE

We perform experiments to compare accuracy between using HyDE or raw query embeddings for dense retrieval.

- HyDE won in 8, raw query in 5, and 1 tie
- No clear trend with respect to k observed
- Adds additional computation cost in generating the hypothetical document

Our results do not lead to a clear conclusion on which method is better. We leave this as a configurable option.

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Analysis of Findings

We find the following issues in analyzing failure cases.

- Multi-column document comprehension errors causes noise in embedding and retrieval
- Knowledge graph option retrieves context using flattened community summaries, loses specific details
- More retrieved text benefits end-to-end generation accuracy, but too much visual context may harm generation
 - Visual retrieval provides fine-grained visual information for answering questions
 - Large batches of images may introduce noise and confuse the MLLM

Limitations

- Preprocessing overhead and limitations
 - Using MLLM for image captioning incurs an overhead of average 3.89 seconds per image
 - Video processing not included in our pipeline
- Dense retrieval theoretical limits
 - Theoretical limits in retrieval effectiveness with respect to the number of documents to store, the number of items to retrieve, and dense vector dimensionality
 - Corporate-grade applications with hundreds to thousands of documents may reach such limits
- MLLM perception lacking behind human vision
 - Pipeline based on assumption that relevant images provide visual evidence to further ground generation
 - Current MLLMs show severe deficits in visual perception without textual fallback
 - Severe lack in MLLM capability when visual content cannot be translated into natural language

Future Work

- Better preprocessing efficiency, such as by parallel MLLM calls for captioning
- Fine-tuning foundation models (DINOv2, Talk2DINO) for specific domain applications
- Retrieval methods: agentic retrieval for better flexibility, rerankers to sort top- k retrieved items
- Further research on improving MLLM visual perception capabilities
- Application to text-to-vision models for better use of retrieved visual details

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Conclusion

We present a multimodal Retrieval-Augmented Generation (RAG) pipeline in this thesis, with a focus on visual elements.

- We provide a complete pipeline that supports text and text-image queries, with evidence covering both modalities;
- We utilize state-of-the-art pretrained foundational models (DINOv2, Talk2DINO), a method without fine-tuning requirements;
- We achieve results on par with or slightly better than existing methods on research benchmarks.

Despite limitations in preprocessing and MLLM perception, our pipeline offers a foundation for grounding MLLM generation with retrieved visual evidence.

Future work will explore better preprocessing efficiency, agentic retrieval and reranking, domain-specific fine-tuning, and extension to application with text-to-vision models.

Thank you

Thank you!

Questions?